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I. Problem Definition

a. Problem Statement

The problem of	members of cultural communities being unable to find culturally specific food products in major grocery chains, while nearby independent stores that carry these products remain unknown to them
Affects	Members of cultural communities; Owners of small independent grocery stores
The impact of which is	<i>Consumers:</i> they don't know that the products they want are sold near where they live, and those without a car or with limited mobility often live far from the neighbourhoods where these shops cluster, so each trip is long and costly. <i>Store owners:</i> their customers are limited to their immediate neighbourhood, their store and its inventory go unnoticed by the wider community, and stock goes unsold.
A successful solution would be	A designated platform for small independent stores letting consumers discover which nearby independent stores carry the culturally specific products they are looking for; bring those products to consumers without requiring long trips, particularly benefiting those without a car or with limited mobility; extend store owners' reach beyond their immediate neighbourhood, making their stores and inventory visible to the wider community; and help store owners sell their stock before it goes to waste.

Disclosure: Used Claude to rephrase the solution benefits in the problem statement.

b. Product Position Statement

For	Members of cultural communities in Montreal
Who	Want culturally specific food products but don't know which stores carry them, or cannot easily reach those stores because of time, distance, or mobility constraints
The "Little Isles"	Delivery platform dedicated to independent grocery stores
That	Lets users discover the full range of products offered by independent stores near them
Unlike	General-purpose delivery apps such as Uber Eats, Skip, and DoorDash, where independent grocers are mixed in with chains and restaurants
Our product	Lists only independent stores, not grocery chains, and lets users filter stores by cultural type

c. Product Overview

Our proposed product would be a delivery service that would allow users to get groceries delivered from stores that don't normally offer that service. Customers will be able to browse from available stores near them, place orders, pay online, and select a delivery time. Stores will be able to manage their offered products and accept and manage orders through a merchant specific interface. Courriers will be able to accept open deliveries and update order statuses. Platform operators will be able to intervene in case of problems with an order or store and manage platform issues as a whole.

d. Product Perspective

Customers, stores, couriers and platform operators will each have their own user interface to interact with the system according to their role. While the platform will be independently operated, it will require communication with third party services to accomplish some functionalities. A payment processor will accept payment from customers and assign payment to the stores and couriers as well as the platform in the form of service fees.

e. Assumptions and Dependencies

Assumptions	Dependencies
Users will have internet connections	Compatible web browser
Stores will update their inventories	Payment processor service
Each order comes from a single store	Hosting and database services
All stores will have a static delivery range	
Stores fulfill the order; couriers only take care of delivery	

II. Technology Planned: *[technology used for the following activities]:*

a. Control Versions System

GitHub (Our team has settled on selecting GitHub as it offers an array of desirable features. Firstly, the repository functionality enables shared access of the web application's code amongst all team members, allows for code to be worked on in various branches, and may store previous working versions of our system alongside relevant documentation. Next, GitHub facilitates the separation of tasks by assigning contributors to their respective issues, of which their specifications are clearly mentioned. Lastly, automation may be achieved via the utilization of a CI/CD pipeline allowing for a seamless transition between development, testing, and deployment of a new feature or version.)

b. Team Collaboration

The team was divided into two smaller teams to create the diagrams. The first team, Yasmine, Thanh-Lam, and Brandon, worked together on the context model, while the second team, Jeremie, Sheryl, and Younes, worked on the domain model. Then, a Discord meeting was scheduled to peer-review each diagram and make sure everyone was on the same page. The Phase I document was a

shared Word document, allowing everyone to collaborate and review each other's work by giving them comments.

c. Design and Modelling Work

Draw.io

The best choice for creating the context diagram and domain model was Draw.io. This web application offers a shared document option, which allows multiple teammates to work on the diagrams at the same time.

d. Development Framework

React

The development framework our team will be using is React. React is a JavaScript frontend framework that will serve as the client-side web application layer for our Grocery Delivery System. This tool will allow us to create our user interfaces such as the grocery catalog, shopping cart, checkout flow, and live order-tracking.

e. Coding

Java

Team DevsOnBevs will be using Java for the backend as it is the recommended object-oriented programming language in the requirements of the project. This will serve to build the server-side logic of our Grocery Delivery System. It will process API requests from the React front-end and securely connect our database.

SQL (SQL Server Manage Studio - SSMS)

To store the data from our users, we will utilize SQL as our database. Via SSMS as our interface to administer our database management system, SQL will handle the continuously changing data of our web application. It will account for the grocery inventory, user profiles, order histories, etc.

f. GenAI Tools Used

In the scope of *Phase I*, generative AI tools were utilized to aid in the realization of the software project. Our team DevsOnBevs ensures the proper use of GenAI by following the guidelines specified in the requirements of our delivery. As such, the delivery is constrained to Level 1 which

is defined by limited AI use and required disclosure. Hence, the LLMs and their role in *Phase I* of our delivery are described in the following.

ChatGPT

ChatGPT was used to brainstorm potential assumptions and dependencies that were left unconsidered. In other words, our team verified that every assumption and dependency crucial to the schematics of our web application were included in the final documentation.

Claude

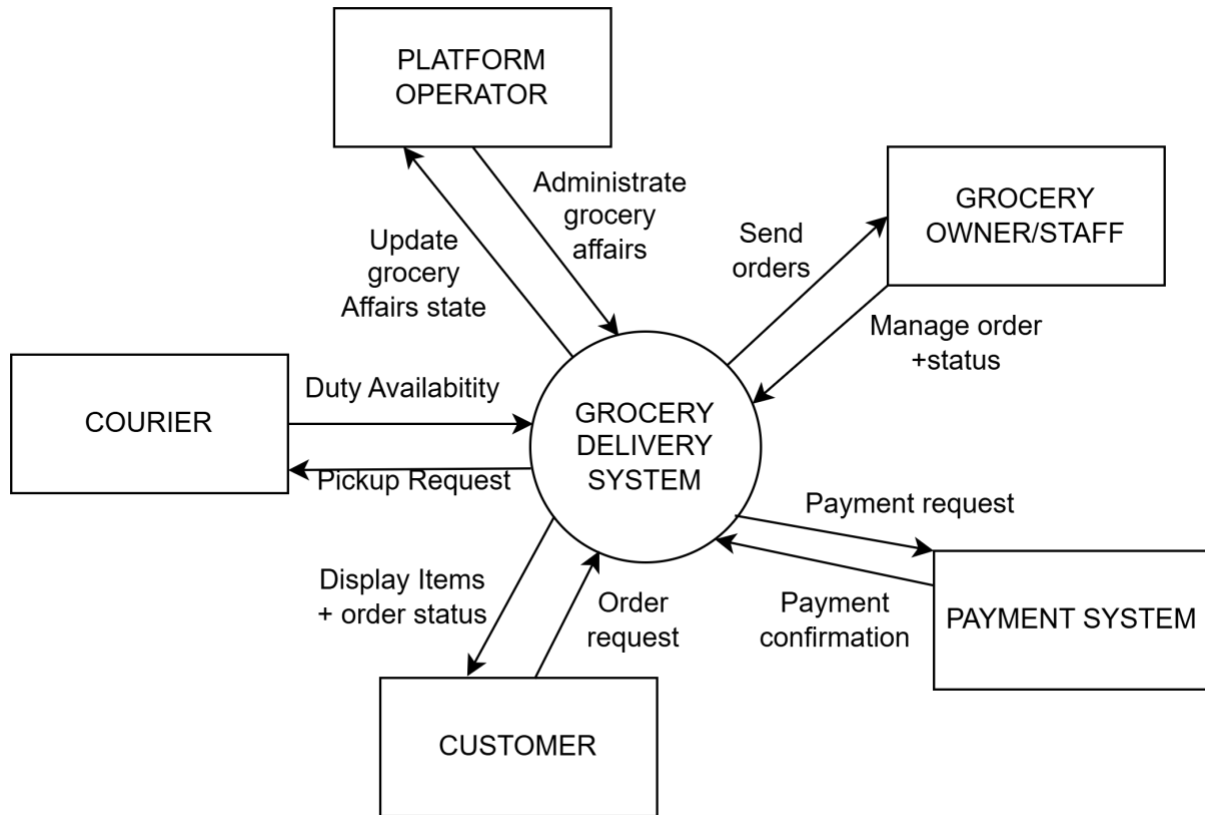
Claude helped concretize ideas regarding the initial food delivery system design by transforming key words into realized sentences. Afterwards, the AI-assisted text was reviewed, edited and approved by all the members of the team.

Claude was also utilized to review the domain model by questioning our hand drawn design to expose the gaps we might have been missing.

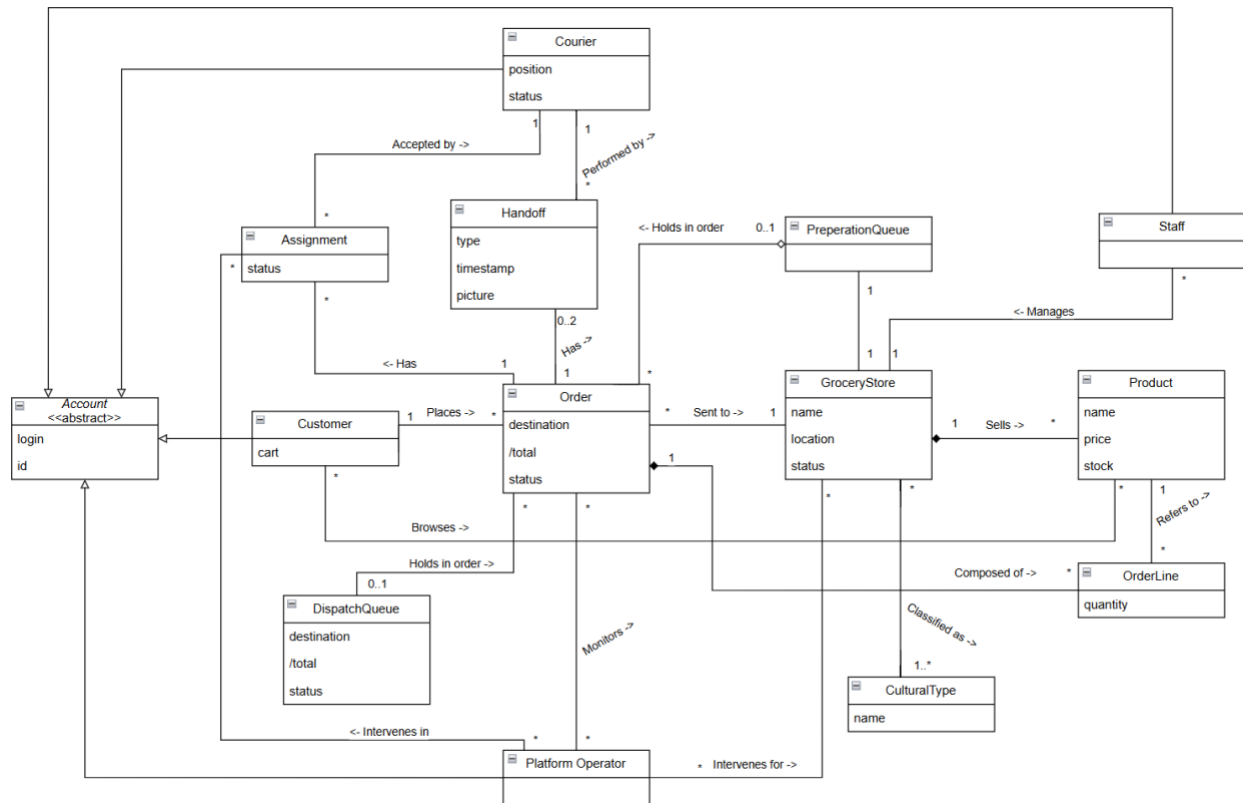
Gemini

Gemini was consulted to answer our questions regarding the structure and guidelines of the context diagram. Our team used these insights to make decisions about the design of the diagram and the software's architecture.

III. Context Diagram



IV. Domain Model



V. References

Sheryl browsed Uber Eats, Skip, and DoorDash to see if they do have the proposed solution in this document.